

ActInf Textbook Group ~ Cohort 2 ~ Meeting 10 (Chapter 5, part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_2 | Cohort 2 ~ Meeting 10 (Chapter 5, part 1) | 55:03

Hello, this is Active Inference Textbook Group, Cohort 2.

We're in Meeting 10 on November 4th, 22, and we're having our first discussion on Chapter 5, Message Passing and Neurobiology.

So, to begin, does anyone want to share any general thoughts or remarks on Chapter 5?

Just any experiences they had reading it?

Any topics that they thought they want to discuss a little bit today?

Just anything about their reading of Chapter 5?

Yes, Ali?

Well, I think in Chapter 5, since the authors decided to go pretty deep into neuroanatomical structures or neuroanatomical substrates of most of the Active Inference concepts, well, having a kind of basic understanding of neuroanatomy, neuroanatomy, in my view, is an essential prerequisite to understanding the contents of this chapter. But otherwise, it provides some empirical evidence for why Active Inference Modeling works, or to put it this way, it explains some of the testable predictions we can get from applying, from seeing Active Inference from a neuroanatomical perspective.

Awesome. Thank you.

Anyone else want to share a general thought on Chapter 5?

Yeah. As Ali stated, the chapter is going to provide a host of examples, which are going to be discussed separately, and then synthesized together, regarding neurobiology, specifically mammalian neurobiology, which is one of the most important systems of interest of us as humans. And also, it's one of the most studied in terms of different zones where Active Inference models have been applied, ranging from, as we'll see in this chapter, the micro circuit level. So, the connectivity patterns of small numbers of computational elements, which might be like neurons, or it could be brain regions. Then, there's a discussion of motor commands in terms of continuous time Active Inference, and how motor behaviors in the periphery are coordinated. There are then several discussions about decision-making, neuromodulation, and learning. And then, there's a hierarchical hybrid model that's presented, which integrates the continuous and the discrete cases of Active Inference in a unified generative model. And that's what brings it all together in the summary. And also, just one note, maybe we've mentioned before, Chapter 1 provides an overview of the book, much like Chapter 10.

Chapter 2 and 3 are the roads to Active Inference, but not Active Inference itself, though it also kind of is.

Chapter 4, we learned about the essence of specifying an Active Inference entity, which is its definition in terms of a generative model.

And now, immediately, we're moving to the area with the most empirical support and popular interest, which is understanding Active Inference in the context of mammalian neuroanatomy.

And that will conclude the first half of the book, which is the epistemic component.

The second half of the book, Chapter 6 through 10, heads into some of the step-by-step and recipes around designing Active Inference models.

But in some ways, as we complete this last chapter in the first half of our cohort, it's just good to keep in mind that we spent 60% of the time getting into the game.

Then, where the low and the high road intersected was in the generative model.

And now we're immediately going to be talking about topics related to applications and specific generative models applied to neurobiological phenomena in health and in disease.

Okay.

Does anyone else want to add a comment on Chapter 5 before we start moving through it?

Okay.

Does anyone else want to add a comment on Chapter 5 before we start moving through it?

All right.

Chapter 4 described the generative model.

In this chapter, there's a focus on the process theories arising from those dynamics, theories that explain how the brain may implement variational inference.

So, as per classical scientific criterion, the goal of a theory or a contribution is to variously explain, predict, design, control.

These are the criteria that active inference can and should be compared and juxtaposed with other competing theories.

And so, many of these aspects are going to be touched upon.

Explain and predict.

Explain and predict.

Explain and predict.

And ultimately, to design and control.

But initially, in this setting, to explain and predict.

So, we'll be keeping an eye out for unique and valuable explanations and predictions that active inference modeling of a given system of interest provides in comparison with some other approach that could be taken.

Let us take a step back from the technical material of Chapter 4, the generative model, and turn our attention to the process theories accompanying active inference.

It's important to draw a distinction between a principle, like the free energy principle, and a process theory about how the principle may be implemented in a certain kind of system, such as the brain.

The latter lets us develop hypotheses that are answerable to empirical data.

So, quite commonly, we hear about the non-falsification or the non-falsifiability of the free energy principle as a principle akin to a principle of least action.

And one can see that comment addressed at the beginning of this chapter, which is, sure, whatever you think about whether free energy principle is tautological or it can or can't be falsified, when we develop specific hypotheses embodied as models of given systems of interest, then we're certainly in the space of being able to find unique explanations and predictions to falsify or not. Like, we could have one model that says that two brain regions are connected in some way, and then there could be an experimental study that falsifies or adds or removes evidence to support that claim.

So, this is where we actually get into the last mile of scientific modeling.

Where we're going to be saying specific things about specific systems.

The dual aim of this chapter is to introduce readers with a technical background to the neurobiology of active inference.

So, that's people coming from a math and modeling perspective to learn about some of the important neurobiological substrates

where active inference models have been constructed over the last 10 or so years.

And to highlight to biologists the relevance of theory to practical neuroscience.

So, for those who are coming from more of an embodied or biological angle here,

why does it matter that we're adding this mathematical layer on top of the raw anatomical observations?

Like, the structural connectivity of different brain regions is an empirical measurement about the world.

So, what are we doing that goes beyond saying this is connected to that?

And the chapter is not the final word.

It is simply some interpretations that seem most consistent with the currently available evidence.

Here, they describe how in the following sections, they're going to bring up first, as mentioned briefly before, the role of the microcircuit as a computational component, which is very common to computational neuroscience.

Then, moving through motor and cortical regions of the mammalian neuroscience to understand first modularly,

and then in composition, how decisions around motor behaviors can be seen as the coordination among multiple active inference generative models.

Anyone want to give a general thought or just like something that's interesting or exciting about this framing?

Okay.

Section 5.2, microcircuits and messages.

So, in Chapter 4, there was a discussion about belief updating,

which we spoke a little bit about in terms of message passing and belief propagation in Bayesian graphical

networks.

And here, they provide some written overview of the layered structure of the mammalian cortex.

So, does anyone want to give a note on the mammalian cortex or the layer structure, the so-called six-layer structure of the mammalian cortex?

Okay.

So, just for context, this is like, let's see where it's actually sliced through.

Here, we have a prototypical mammalian brain.

I'm always emphasizing mammalian because this is not some general property of brains.

Like, invertebrate brains have a very different layout.

They don't have cortical layers in the same way.

And so, these cortical regions near the top of the head,

if one slices through them,

they reveal a very specific pattern of cells

that are arranged in these laminar or layer-like ways.

And so, there are these six areas.

Six layers.

One, two, three, four, five.

Maybe that should be six.

Maybe there's some other thing happening here.

But this is representing like a slice through cortex.

And so, this has been modeled as a column.

Because as you go across the cortical surface,

there are these repeating units that have this kind of laminar structure.

It's like six layers of lacquer across the cortical surface

once it's unfolded and sliced through.

And that's represented from a Bayesian computational view in figure 5.1.

Canonical cortical microcircuits illustrating relationship

between inferential message passing

and the architecture of the cerebral cortex.

On the left is a schematic of some anatomy

where the different connections represent inhibitory or excitatory synapses.

So, inhibitory synapses reduce the likelihood

of the downstream neuron firing when activated

So, the left is not a Bayes' graph.

It's actually a summary of structural, anatomical, histological,

that means tissue level anatomy, connectivity.

In the middle is a message passing architecture.

So, like a scheme for variables and their connectivity

within a hierarchical predictive processing architecture.

And on the right side is a message passing scheme

that's described to solve a partially observable Markov decision process, POMDP.

Now, on the left is an empirical, neuroanatomical, structural map.

On the right, we see two computational implementations of inference and action.

And they're being kind of laid out.

They could be laid out in any way.

They could look like the Big Dipper graphically.

But they're being juxtaposed with the cortical layers

so that we can start to see some concordances

and resonances between the location of a given variable

in a statistical model of inference and action

and the neurobiological substrates

who, with plausible biological connectivity,

might be implementing or doing something like

the computational role that the variables play

in the abstract models.

So that's going to be the path towards,

or the template of the path,

towards unique explanations and predictions

from using computational modeling

of the structural neuroanatomy.

Does anyone have any thoughts or questions on

figure 5.1

or like how a structural anatomical empirical map

might be juxtaposed with statistical models?

Maybe a good thing to think about

or question and answer here is like

what connections or what structure

reflects something like a belief

in the statistical sense?

Or what connection could be made there?

So what neurobiological structures

can reflect or represent beliefs?

Sure.

Yeah.

That's fair.

Or in what way are they different?

Or, yeah.

What's not a belief here?

Does anyone have any...

A lot of errors.

Thoughts?

Yeah.

And then the quarterback

is going to pass it to you.

Yeah.

One thing I noticed immediately

in this previous...

It was just the cortical layers there.

And the...

There's this 2

on the right-hand side of that

for layer 3 and layer 4.

So I'm not sure what...

exactly...

Yes.

Oh, okay.

That's I.

Yep.

Yep.

One important thing to note

about the cortical layers,

which is almost

obtrusively under-discussed,

is the cortical column

performs as one integrated unit.

The fact that it's 6 layers

doesn't mean that there's

like a little actin loop

and then a second actin loop

and then a third actin loop.

So it's not like a 6...

If someone said,

I'm looking at a 6-layer

predictive processing hierarchy,

one might think that there's
like 6 cleanly separated layers
of a wedding cake.
But actually,
the cortical column
has 6 arbitrarily defined.
Not to say that there's no grounding,
but there's finer scale differences
and there's coursing
and there's overlapping zones.
And it's behaving with connections
that span across multiple layers.
So this isn't like a clean,
6-layer predictive processing model.
This is like one integrated architecture
that functions with this connectivity.
And importantly,
this is the basis
of like the thousand brains hypothesis
and Numenta AI
and people who have been pursuing
the cortical architecture
as the or a basis
for intelligence
in natural and artificial systems.
And it's long been hypothesized
when comparing the brains
of different mammals,
like what neuroanatomical features
are associated with differences
in perceived intelligence
or learning capacity.
And one of the features
is the prevalence
in the primates
and in humans
of higher cortical regions,
so-called higher cortical regions

with this specific architecture.

Whereas other brain regions

maybe have not

had so much evolution

in their size

or in their circuitry.

that doesn't mean

that they're

unimplicated

in intelligence.

It just means

that people have looked

for where there are

more differences.

To this question

that Brock asked

about like

where might beliefs

be embodied.

So within the

Bayesian setting,

the statistical setting,

a belief

is just any

parameterization.

can be said

to be a belief

about a variable.

So we're not

talking about

experienced

or personally held

or verbally

statable beliefs.

We're talking about

statistical parameterization

of a given variable.

And there's

at least two ways
one could think
about beliefs
being represented
or embodied
in a structure of this.

And we're going
through this
in slow time
because like
this kind of
general questions
about what is
the structural
neuroanatomy
and how does
it relate
to statistical
modeling thereof
is going to be
at the heart
of the following
examples
as well,
not just this
cortical
case.

One can see
the firing rate
of a neuron
in like an
artificial neural
network setting
or just
the value
that a statistical
variable takes on
as representing

a belief
about that
variable.
Then
one can think
about the
presence or absence
or the weighting
or the type
like inhibitory
or excitatory
of a connection
between two
variables
as embodying
a hypothesis
about
the structure
of how those
variables are
connected.
under the
implicit
assumption
that we're
developing
models that
actually do
functional things.
Of course,
one could just
connect all of
the variables
and have an
all-by-all
model.
However,
as we

sparsify the
relationships
amongst variables,
which is to
say remove
edges from the
all-by-all,
the model
becomes computationally
easier to fit
and we
talk about
that in
terms of
the
factorizability
of the
variational
Bayesian
approach.

So if
none of
the variables
were connected,
they're all
independently
fit.

That's like
the simplest
it can be
to fit
 n variables.

If all
the variables
were interacting
in combinatoric
ways,
that would

be increasingly
computationally
complex.

And for a
given amount
of data,
we would
have less
and less
statistical
power to
resolve any
given variable
or combination
of variables
because the
same amount
of information
from our
observations
is now being
like partitioned
and spread
out over
fitting estimates
for more and
more variables.

And so
what biological
systems do
and what
statistical
models do
is enforce
either just
through physical
constraints
or through

design
sparsity
conditions
on
functional
statistical
modules
such that
the models
are more
computable.
Though they
only may be
exploring a
subset of
the space
of total
possible models,
exploring well
a subset
whose sparsity
embodies
certain aspects
of the world
may be a
more appropriate
model to
fit
situationally.

figure 5.2
simplified
version of
the predictive
coding scheme
shown in the
middle of

5.1

So does

anyone want

to give a

thought on

5.2

So here's

message passing

between three

cortical regions.

So here's

three cortical

columns,

each consisting

of like one

through six

layers.

And

there's going

to be message

passing

happening

laterally.

dotted lines

show ascending

messages,

prediction errors,

and solid lines

show descending

messages or

predictions.

So in this

setting,

this rightmost

region is

like the

quote top

most,

but left,

right,

top down,
bottom up,
those are all
just spatial
metaphors.
But we'll
continue to
use this
higher and
lower in
the hierarchical
modeling where
the lower
levels of
the hierarchical
model are
more close
to where the
rubber hits
the road in
terms of like
the perceptual
machinery and
the action
effectors.
And higher
levels of a
hierarchy are
associated with
more and more
abstract or
general or
deeper or
slower features
of the world.
So this
figure is
representing

how there
is a
within
cortical
column
relationship
between
predictions
and prediction
errors,
between the
superficial and
the deep
levels
using the
concordance
between
variables and
cortical
layers established
in 5.1.

And then
the
outcome
of one
cortical
layer
can also
be engaged
in this
prediction
and prediction
error
relationship
with
an
anatomically
lateral

cortical

column.

However,

from a

computational

perspective,

this is

like the

top,

the middle,

and the

bottom

of a

three-layer

cognitive

model.

The

superscripts

are describing

the level.

So here

is like

i ,

i plus

one,

and i

minus

one.

So this

is just

describing

that this

is a

scheme

that can

be

computationally

extended to

many levels
of hierarchical
modeling.
and it's
being embodied
by the
lateral
juxtaposition
neuroanatomically
of cortical
columns.

Ali, did
you have
something you
wanted to
add there?

Well,
yeah,
I just
wanted to
mention
Friston's
recent paper
from a
couple of
months ago
in which
the
neurobiological
substrate
of
prediction
errors
are
explained
in
detail,
namely,

the
computational
psychiatry,
which is
a great
overview
paper
of the
whole field
of computation
psychiatry
through

lens
of
active
inference
and
especially
in
on the
matter
of
prediction
error,
I'm sorry,
a matter of
prediction
error,
I just
wanted to
point to
the section
and
let me
sorry,
yeah,
here are the

importance
of being
precise.
I just
wanted to
quote a
couple of
sentences
from this
paper,
which says
predicting
or estimating
precision
is a
universal
requisite
for making
sense of
data,
from
estimating
the
signal-to-noise
ratio
in some
sensory
apparatus
through to
estimating
standard error
when making
inference
bias on
Nyman-Pearson
statistic.
Affording
certain

prediction

errors,

greater

precision

increases

their

influence

on belief

updating

and has

all the

hallmarks

of

attentional

selection.

Physiologically,

this simply

entails

an increase

in the

excitability

or

post-synaptic

gain of

neuronal

populations

broadcasting

prediction

errors.

In this

view,

there is an

intimate

relationship

between

attention

and the

modulation

of
synaptic
efficacy
by
classical
neuromodulators
and
non-linear
post-synaptic
responses
responsible
for
mediating
the
exchange
between
fast
spiking
inhibitory
interneurons
and
superficial
pyramidal
cells.

So
it's
basically
the
arguments
from
the
book
but
in a
much
more
concise
and

a
little
bit
well
structured
argument
in my
opinion.
So
if
anyone
wants
to
just
look at
this
paper
I
think
it
might
be
helpful
in
grasping
the
material
which
is
I'm
posting
the
link
here
in
the
chat.
Awesome.

Yeah.

Thank you for
sharing it.

Just to
kind of
again as
we walk
towards
unique
explanations
and
predictions
for
active
inference
models
of
empirical
neuroanatomy
we can
think about
at least
a few
areas
of
alignment
between
these
models.

So
there's
structural
alignment
where
realized
anatomical
connections
have

parallels
to
statistical
connections
in a
Bayesian
graph.
they are
not one
and the
same
but
it's
just
an
area
of
alignment
that
can
be
used
to
explore
predictions
and
explanations.
Then also
there's
two areas
that are
mentioned
by what
Ali just
read
which are
some of
the

functional
components
for example
neural
firing rates
or
synchronizations
and other
sorts of
dynamical
functional
relationships
like
when this
variable
is high
that
one
is
lower
whether
they're
connected
or not
structurally
and
chemical
features
like
neuromodulation
or the
presence
or absence
of an
excitatory
or
inhibitory
neuromodulator

at a
given
synapse
or even
taking us
into
synaptic
plasticity
and the
neuromodulation
of
Hebbian
associative
learning
all of
these
areas
when
brought
together
in a
total
fabric
of
evidence
help us
strengthen
the
relationship
between
the
computational
models
and the
neuroanatomical
empirical
findings
and then

they suggest
things to
each other
So that
was heavily
on the
cortical
architecture
of the
column
and then
how
laterally
linked
columns
which again
is their
empirical
structural
connectivity
patterns
can be
thought of
in terms
of
hierarchical
Bayesian
modeling
then
they move
into
motor
commands
So
let's
think
about
the

layer
V
of
the
cortex
projecting
as a
projection
So
layer
V
of
the
column
here
is
this
DP
deep
pyramidal

That's
just
the
name
of
the
cell
it
has
a
certain
architecture
that
makes
it
called
pyramidal

This is
like
classic
Santiago
Ramon
Cajal
stuff
and
it
has
this
arrow
coming
out
going
to
motor
neurons
So
here's
like
a
motor
output
from
this
cortical
column
Not
all
cortical
columns
are
involved
in
motor
outputs
but

some
are
So
we're
going
to
be
receiving
inputs
from
those
pyramidal

Neurons
and
it's
going
to
be
projecting
onto
a
butterfly
No
this
is
a
cross
section
of
the
neural
cord
the
spinal
cord
of
a

mammal

Figure

5-3

Neuroanatomy

associated

with

active

inference

and

modulating

spinal

motor

reflexes

Coming

in

to

the

system

from

the

proprio

sensors

afferent

meaning

like

away

from

the

sensor

as

opposed

to

efferent

The

proprioceptive

afferent

signal

is being

compared
in an
error-like
fashion
with an
epsilon
with
the
descending
top-down
predictions
coming from
that
cortical
column
that we
just
looked at
The
output
of
this
comparison
is
a
motor
message
or
command
or
signal
to
a
motor
actuator
So
one can
imagine

that where
proprioception
exactly
matches
expectations
around
proprioception
no
motor
movement
might be
passed
forward
I'm
trying to
hold my
hand
on the
table
all
signals
are
suggesting
that the
expectation
is being
realized
and no
movement
is
enacted
In
comparison
if one
expects
to be
receiving
different

proprioceptive

afferent

information

which is

to say

to be

in a

different

postural

position

then

motor

commands

can be

enacted

to

reduce

that

divergence

So

different

proprioceptors

for example

are in

your

elbow

such that

there's a

different

proprioceptive

afferent

coming in

when your

elbow is

extended

and when

it's

completely

closed
and then
expectations
in the
head
the brain
can
change
such that
different
motor
commands
are
enacted
to
open
and or
close
the
elbow
dynamically
as
authentic
proprioceptive
information
comes in
So
broadly
that is
this
neuroanatomy
Any
thoughts or
questions
on it?
Yes Ali
Yeah
Actually

there's
I think
it was
in the
beginning
of
section
5.3
there's
a sentence
which
might
result
in some
misunderstanding
of confusions
which says
the schematic
on the left
in figure
5.1
shows that
layer 5
of the
cortex
projects
to
spinal
pyramidal
neurons
and that
this can
be interpreted
as a
prediction
but if
we look
at

figure

5.2

5.1

we won't

see

this

explicitly

I mean

anything

labeled as

spinal

pyramidal

neurons

because

because

the primary

excitation

neural cell

types

in corticospinal

tract

is

pyramidal

neuron

spinal

motor

neurons

and

spinal

parameter

neurons

are sometimes

used

interchangeably

so

in this

figure

actually

what we
see
is
an
excitatory
connection
which is
drawn
from
layer
5
to
spinal
motor
neurons
and
not
spinal
pyramidal
neurons
but
they're
basically
the same
thing
so
I just
wanted
to
clear up
that
if
anyone
has
that
misunderstanding
thank
you

the
pyramidal
neuron
there's
a large
diversity
of
structure
but
if
you like
squiggles
that look
like this
you'll
love
cellular
neuroscience
alright
there's
an
extremely
interesting
discussion
around
sensory
attenuation
in the
context
of
motor
initiation
and
completion
so
we need
a way
when we

want to
make a
move
so to
speak
we need
a way
to preclude
sensory
data from
updating
our
expectations
so that
we can
entertain
the
initially
false
belief
that we
are
moving
so
this
belief
can
be
realized
through
action
so
let's
just
imagine
that
we're
in

a
case
where
we
are
expecting
the
elbow
to be
closed
but
the
proprioceptive
information
is
suggesting
that
it's
extended
if
you know
the
cortical
hierarchy
simply
changes
its
belief
to
I
believe
that
it
is
closed
that
will
immediately

be
overridden
by the
veridical
proprioceptive
information
coming in
and
so
as we
know
there's
two
ways
to
update
the
generative
model
and
to
minimize
free
energy
changing
the
world
and
changing
your
mind
so
here
it's
like
change
the
body

or
change
the
mind
and
under
high
precision
information
coming
in
changing
one's
mind
will be
brought
back
into
alignment
with
the
high
attention
which is
to say
high
precision
to the
sensory
information
coming
back
in
and
so
the
computational
strategy

implemented

is

that

the

descending

motor

tracks

predict

not

just

the

data

but

the

precision

or

the

confidence

placed

in

those

data

so

high

precision

high

confidence

high

attention

that

means

you

snap

to

whatever

that

is

being

whatever
that
high
attention
high
confidence
high
attention
variable
is
saying
in
the
extreme
just
whatever
it
said
in
the
last
time
point
becomes
your
estimate
the
opposite
extreme
is
low
confidence
low
precision
low
attention
and
in

that
case
no
matter
what
that
signal
says
you
don't
change
your
mind
at
all
and
so
what
happens
is
previously
we were
talking
about
the
elbows
prediction
around
being
open
or
closed
but
now
let's
think
about
it

as
providing
an
expectation
and
variance
coming
down
so
initially
the
confidence
placed
in
those
data
are
decreased
that
allows
movement
to be
initiated
this
is known
as
sensory
attenuation
and can
be thought
of as
the
complement
to
sensory
attention
so
attenuation

is low
attention
equipping
us
with the
capacity
to
ignore
certain
prediction
errors
such as
generated
by
saccadic
eye
movements
where
sensory
attenuation
is known
as
saccadic
suppression
so
other than
the blind
spot
and so
on
saccadic
suppression
is one
of the
most
interesting
examples
of

embodied
precision
modulation
like
when you
do an
eye
saccade
which can
happen
multiple
times
per
second
there's
no
nausea
or vertigo
except
when it's
pathological
because
during that
motor
movement
there's
a reduced
attention
to the
bottom-up
visual
input
and this
is
highly
empirically
demonstrated
in multiple

areas of
the visual
processing
neurobiology
so that
like
when the
eye
is
still
there's
an
increased
attention
to the
incoming
raw
bottom-up
information
when the
eye
is moving
there's
a decreased
attention
to the
bottom-up
information
and an
increased
reliance
on the
generative
model
during
those
transients
of

saccade
and so
similarly
to initiate
a movement
requires
an initial
suppression
of
precision
alongside
a change
in
expectation
of
proprioception
and that's
what allows
one to
believe
like okay
I'm gonna
get out
of the
chair
I believe
that I'm
out of
the chair
and then
that shifts
the balance
of minimizing
!
free energy
towards
changing
your body

aka changing
the world
through action
rather than
immediately
just getting
your
change your
mind
knocked
back
into
place
and
interestingly
this is
absolutely
concordant
with many
healthy
neurodiversity
and pathological
scenarios
involving
the
neurobiology
of
failure
to
initiate
continue
or terminate
movement
movement
so like
there's
scenarios
where

somebody
may have
trouble
beginning
movement
but once
they've
begun
movement
they can
walk
normally
alternatively
there are
motor
behaviors
where somebody
may have
trouble
terminating
behavior
and so
these can
be modeled
from a
computational
perspective
so that
not only
are we
able to
say
well
this
chemical
is
higher
in

this
condition
or
people
with
this
diagnosis
tend
to
have
this
comorbidity
but
we can
actually
think
about
the
parameterization
of these
computational
models
and have
some more
nuance
around
well
should
we be
increasing
what
or
modulating
what
Ali
actually
there's
an

interesting
argument
in
Mark
Sohn's
book
The Hidden
Spring
which says
that
because
consciousness
I mean
it says
that consciousness
is only
required to
mediate
mismatches
between
internal
and external
states
the more
adaptively
and
predictively
reliable
our model
becomes
the more
our behavior
becomes
automated
operating
below
the level
of

conscious
experience
so
basically
it says
that
consciousness
is needed
in situations
where
surprisal
needs
to be
minimized
when there's
no
surprisal
there's
not
much
need
for
consciousness
or
in other
words
consciousness
is a
process
that
somehow
seeks
to
overcome
itself
hmm
well
very

deep
thank you
for
sharing
it
okay
just
to
peer
ahead
into
where
we're
going
in the
second
half
of
five
and
then
people
can
think
about
some
questions
hopefully
write them
down
as they
see
fit
so
now
we're
coming
back

into
the
brain
into
the
sub
cortical
structures
so
beneath
the
layer
of
cortex
that
has
the
histology
described
by the
columnar
architecture
are
the
sub
cortical
structures
sometimes
referred to
as being
in
the
midbrain
specifically
there's
going to
be a
discussion

of
the
basal
ganglia
and
colleagues
this is
one of
the
regions
in
the
brain
that
has
been
implicated
through
many
neuroimaging
and
disease
related
and
all
kinds
of
different
studies
in
decision
making
and
learning
and
it's
known
to

have
some
dopaminergic
characteristics
expression
of
signaling
related
to
dopamine
and
actually
only
a
very
small
fraction
of
cells
in
the
mammalian
or
invertebrate
neuroanatomy
are
dopaminergic
in this
way
it's not
like
every
neuron
is
responsive
to
dopamine
some

of the
details
are
described
some
of the
mathematical
parallels
are
drawn
and
here
it
culminates
in
a
indirect
pathway
and
direct
pathway
of
decision
making
that
are
which
choice
between
these
two
pathways
is
favored
is
going
to be
computationally

assigned
to the
role
of this
gamma
variable
and
these
two
pathways
can be
thought
of as
deliberative
decision
making
in the
direct
pathway
and
instinctual
or habitual
decision
making
in the
indirect
pathway
how do
we see
that playing
out
observations
are coming
in
and
in both
cases
policy

pi
are coming
out
so
this
neuro
anatomy
is
implementing
a
computation
that
takes
in
some
type
of
input
from
the
cerebral
cortex
which
again
might
be
a
prediction
about
something
else
and
results
in
policy
evaluation
and
selection

e
is the
variable
representing
affordances
and it
can be
thought of
as a
habit
vector
because
it
characterizes
not just
what can
be done
but the
baseline
frequencies
or probabilities
of those
actions
to be
done
whereas
g
is expected
free
energy
and that
relies
not on
the past
habits
and ratios
of performance
of different

behaviors
without
consideration
to the
impact
of those
behaviors
but
rather
the
expected
free
energy
calculations
around
different
policies
that could
be
undertaken
so
here
we can
start to
see
differences
in
decision
making
where
sometimes
this is
sort of
a type
1
type
2
manifold

in terms
of like
quick
simple
decisions
that don't
even need
to reference
the future
on the
indirect
pathway
and
deliberative
potentially
planning
like
operations
involving
expected
free
energy
calculations
and
a
dopaminergic
neuromodulatory
feature
that
balances
the relative
contributions
of
these two
decision-making
pathways
with respect
to their

influence
on policy
table
5.1
describes
neurotransmitters
their
associated
computational
role
even though
they all
play many
roles
this is
just a
course
graining
and a
sort of
starting
point
and then
empirical
evidence
around
the role
of that
neurotransmitter
in that
computational
way
many
work by
Frishton et
all
many
work

just in
the
normal
neuroscience
realm
5.6
is going
to
move us
towards
it's a
short
section
moving
us
towards
thinking
about
stitching
and
composing
together
hierarchical
active
inference
generative
models
that have
continuous
and
discrete
state
spaces
together
because
as they
write
our

interface
in the
world
around
us
is in
the
continuous
domain
the
implication
of
which
is
that
the
lowest
level
of
any
hierarchy
in
the
brain
must
be
continuous
however
as
decision
making
becomes
more
and
more
abstracted
it
can

be
seen
increasingly
in
certain
cases
as
symbolic
or
discrete
so
from
the
perspective
of a
generative
model
that
means
associating
alternative
discrete
hypotheses
about
the
world
with
continuous
dynamics
entailed
by
those
hypotheses
turn
left
or
turn
right

bring
an
umbrella
don't
bring
an
umbrella
is
it
raining
or
not
friend
or
enemy
juxtaposed
with the
continuous
information
arising
from the
body
and
in
in
summary
with a
short
section
again
they
are
now
going
to
link
together
the

three
neuroanatomical
systems
that are
described
in the
chapter
under
the
spirit
of
this
type
of
continuous
discrete
hybrid
generative
modeling
approach
specifically
we see
this
cortical
hierarchy
that is
engaged
in
a
relationship
in
with the
subcortical
structures
a biologically
plausible
relationship
which reflects

the actual
structural
anatomical
connectivity
between
cortical
and subcortical
structures
and of
appropriate
computational
parallelisms
hashtag
explanations
and
predictions
and
in
the
brain
the
cortical
and
subcortical
architectures
are
ultimately
resulting
in this
study
with
outputs
that
then engage
the spinal
column
in the
reflex

arc
that was
described
earlier
so
policies
might be
discrete
throw the
baseball
or
don't
and
that
can
be
using
complex
cortical
predictive
processing
like
decision
making
facilitated
by this
dopaminergically
mediated
handoff
between
more
instinctual
habitual
like
behavior
and more
deliberative
like

expected
free
energy
comparisons
and
that
can
be
enacted
through
a
prediction
minimizing
approach
carried
out
in the
spinal
column
this
is
continuous
whereas
policies
can
be
discretized
that's
a short
chapter
there's
no
equations
there's
some
variables
in the
figures

but
there
aren't
really
equations
this
chapter
over the
years
will be
fractally
unpacked
to the
most
incredible
level
of detail
they are
just
raising
some
key
points
and
pointing
some
initial
steps
Ali
yeah
as I
mentioned
in the
previous
cohort
I just
wanted to
recommend

this
talk
from
Thomas
Parr
which
summarizes
most of
the
points
in this
chapter
and
some
more
and
I
highly
highly
recommend
it
for
anyone
reading
this
chapter
because
it
really
helps
understanding
some
of
the
more
obscure
concepts
here

thank

you

excellent

okay

any

final

thoughts

or

we

will

close

this

session

okay

thank

you

cohort

two